

LECTURE 1

LECTURER: SAGDOLDANOVA A.A.

ICT ROLE IN KEY SECTORS OF DEVELOPMENT OF SOCIETY





LECTURE CONTENT

- What are ICTs mean?
- Role of ICT
- Difference between ICT and IT
- ICT literacy
- ICT in Education
- Standards in the field of ICT

What are ICTs mean?

ICT stands for «Information and Communication Technologies»

ICT refers to technologies that provide access to information through telecommunications.

For example, social networking websites like Facebook allow users from all over the world to remain in contact and communicate on a regular basis



Definitions of ICTs vary widely, depending on contexts and conditions of use. Here some examples:

TECHTARGET

An umbrella term that includes any communication device or application, encompassing: radio, television, cellular phones, computer, hardware, software and so on

EUROPEAN COMMISSION

The importance of ICTs lies less in the technology itself than in its ability to create greater access to information and communication in underserved populations.

UNDP

ICTs are basically information handling tools—a varied set of goods, applications and services that are used to produce, store, process, distribute and exchange information.

THE ROLE OF ICT IN VARIOUS SECTORS



Agriculture



Education



Healthcare



Business



Transportation



Entertainment



Construction



Military

DIFFERENCE BETWEEN ICT AND IT



In some parts of the world, ICT is used as a synonym for information technology (IT), but the two terms can have slightly different meanings when used in different contexts.

Information technology, or IT, is all about the collecting, processing and sharing of information. The field of IT deals largely with the methods and tools used in said collecting, processing and sharing. These methods and tools include coding and programming, data conversion, means of storage and retrieval, and system analysis and design, among others.

When someone says they work in IT, they're name-dropping a broad field that can include a wide range of professions, including *network administration, computer support, computer science, system analysis, programming* and more. In its modern usage, IT often specifically refers to the information technology that keeps businesses running and connected such as internal servers, web marketing and business software, which IT experts work to develop and maintain.

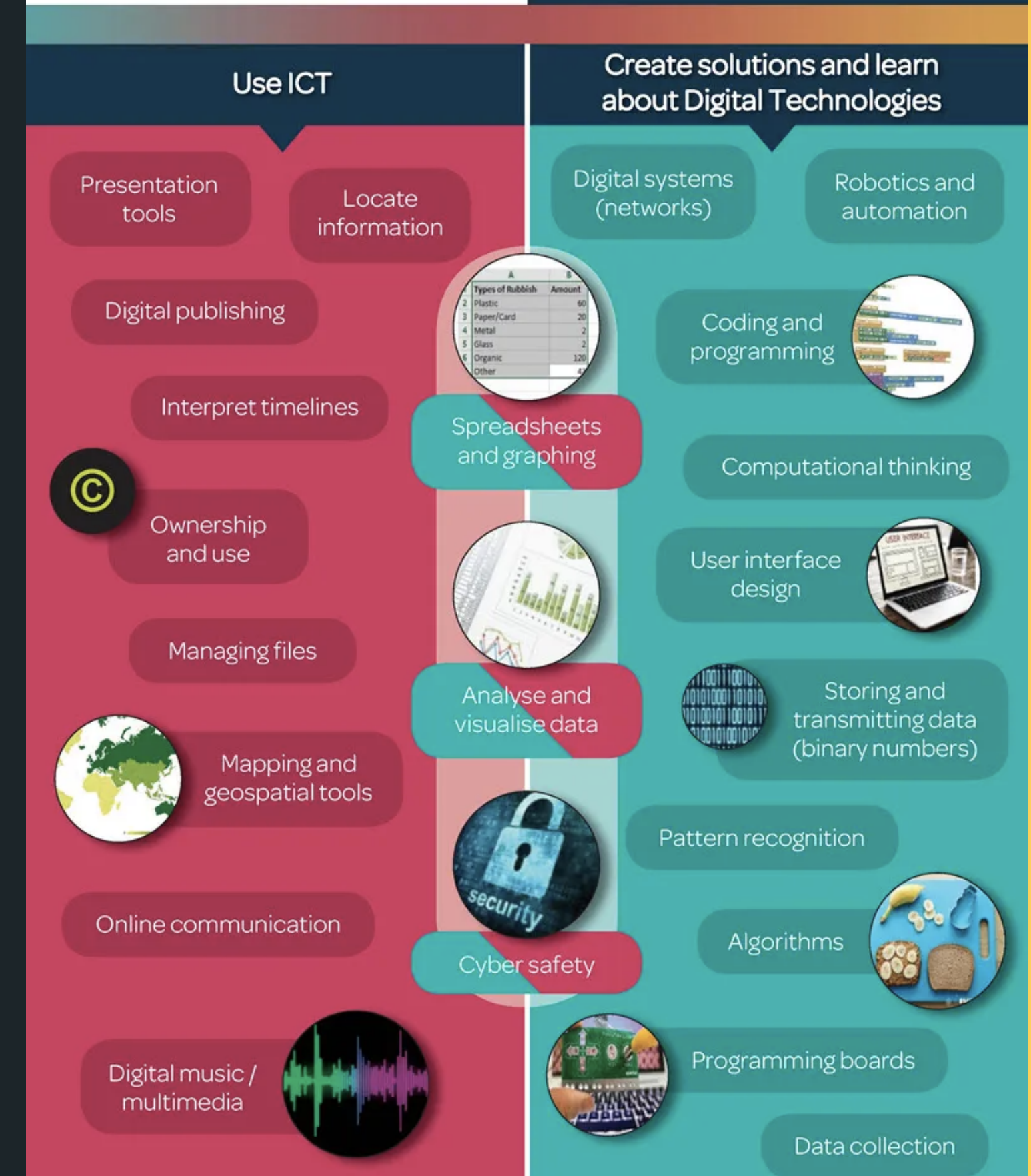
DIFFERENCE BETWEEN ICT AND IT

In the more general sense, ICT is described as using computers and other digital technologies to aid individuals or institutions in handling or using information.

ICT employees can expect to work in positions where the focus of their job includes improving digital literacy, data literacy and using emerging technologies to optimize legacy communication channels.



What's the difference between ICT Capability and Digital Technologies?



DIFFERENCE BETWEEN ICT AND IT

Presentation tools



Digital publishing



File management



Mapping and geospatial tools



Online communication

Digital music / multimedia

Digital systems

Robotics and automation

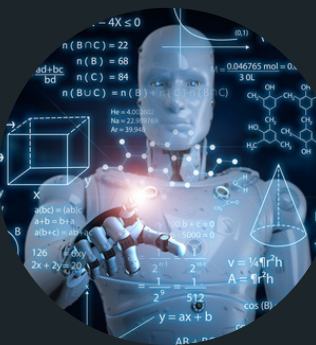
Coding and programming

User interface design

Storing and transmitting data

Algorithms

Data collection





ICT LITERACY

ICT literacy, also referred to as data literacy, focuses on two important things: helping people translate data into meaningful information and communicate that meaningful information to others.

In education and human resource management, ICT literacy is an important skill for employees and students in both STEM and non-STEM fields.

1. Uncertainty	2. Evolving and Experimenting	3. Enlightenment	4. Certainty	5. Data Fluency
Organizations are unaware of the need for data literacy skills and have no or very vague understanding on what is required. Individuals might have a certain interest in data and work digitally but are unsure about the different steps that exist when working with data.	Organizations are experimenting with the use of data to form research questions and address ad hoc needs. A limited number of skilled individuals do most of the work. Many decisions are still made with limited input from available data.	Organizations are expanding the ways in which data can inform their work. Data skills are an increasingly important requirement for staff. Describes a state where a lot about data has already been understood theoretically but cannot be applied in many cases and must be further trained.	Organizations perform data handling steps with confidence and have built data-driven activities into their routine processes wherever it makes sense. Generic procedures and standards on how to handle data are formalized and widespread, benefits are understood at all levels of the organization.	Organizations have established a data-informed culture throughout all levels. Data is actively used to improve processes and create workflows.

Advantages & Disadvantages of ICT

Utility

Shared resources

Digitalization of education, healthcare, etc.

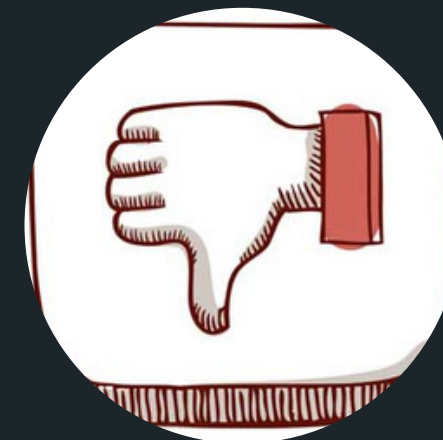
Control

User competence

Vulnerability

Viruses

Complexity



THE MAIN GOAL OF ICT IN EDUCATION



Over the decades, ICT have gradually begun to play a very important role in education, with their usage in education is growing worldwide continuously. These methods allow access to information through the internet. In education, ICT is the method of teaching that uses information and communication technology to support, improve and optimize the transfer of information, in turn leading to the invention of smart education. This improves the quality of teaching, the learning process of students and ultimately facilitates e-learning. It is commonly believed that technology can empower educators and students, making substantial contributions to knowledge and achievement.

ICT in Education



Schools/universities use a diverse set of ICT tools to communicate, create, disseminate, store, and manage information. In some contexts, ICT has also become integral to the teaching-learning interaction, through such approaches as replacing chalkboards with interactive digital whiteboards, using students' own smartphones or other devices for learning during class time, and the “flipped classroom” model where students watch lectures at home on the computer and use classroom time for more interactive exercises



Changes in education due to ICTs



**FACE TO FACE LEARNING
VS
ONLINE LEARNING**

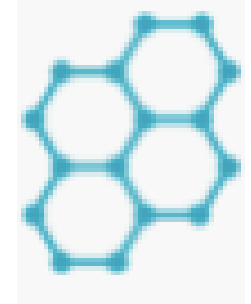
**BLAKBOARD
VS
INTERACTIVE BOARD**



**PRINTED PAPER VS
ONLINE COLLABORATION
AND CO-CREATION TOOLS**



AN EFFECTIVE PRACTICE OF ICT TOOLS IN HIGHER EDUCATION TO IMPROVE THE TEACHING LEARNING PROCESS

Tools	Google Forms	Moodle	Quibblo	Socrative	QuickLearn
logo					
Free/Open Source	Yes	Yes	Yes	Yes	NO
Free Users	Yes	Yes	Yes	Yes	NO
Custom Analytics	Yes	Yes	Yes	Yes	Yes
e-Learning Management	NO	Yes	NO	Yes	NO
Images	Yes	Yes	Yes	Yes	Yes
Videos	Yes	Yes	Yes	Yes	Yes
Mobile Support	Yes	Yes	Yes	Yes	Yes

ICT TOOLS IN EDUCATION

ICT TOOLS FOR QUIZZING/TESTING/GAMING

- BookWidgets
- Gimkit
- Classtools
- Genial.ly
- Google forms
- Classmarker
- Flipgrid
- Deck.toys
- Formative
- Educaplay
- Flubaroo
- Gynzy
- Plickerss
- MasteryConnect
- Lightsail
- H5P
- Kahoot
- JeopardyLabs

ICT TOOL FOR PRESENTATION

- SlideShare
- Buncee
- Prezi
- Prowise
- Google slides
- Nearpod
- Peardeck
- Sutori
- Lesson up
- Mentimeter

ICT TOOLS FOR CREATIVE CREATIONS

- Infogram
- Storybird
- BlendedPlay
- Pixton
- Vizualize
- Wordle
- ScribbleMap
- Canva
- Visme.co
- Easel.ly
- Venngage
- Piktochart

ICT TOOLS FOR ONLINE COLLABORATION

- Asana
- Google spreadsheets
- MeisterTask
- Stoodle
- Edmodo
- Seesaw
- Google Docs
- Talky
- Google Slides

ICT TOOLS FOR VIDEOS/CARTOONS

- Animaker
- DoInk
- Masher
- Binumi
- EDpuzzla
- Educreations
- Biteable
- Clipchamp
- GoNoodle
- ChatterPix

ICT TOOLS FOR BRAINSTORM & ORGANIZING

- Coggle
- Mindomo
- Answer garden
- Postermyswall
- Popplet
- ChartGo
- Xmind
- Timetoast
- Thinglink
- Timeglinder
- Note App
- Trello

ICT TOOLS FOR STUDYING

- Cram
- WRTS
- Cobocards
- Vocabulary
- Learningpod
- Studyblue